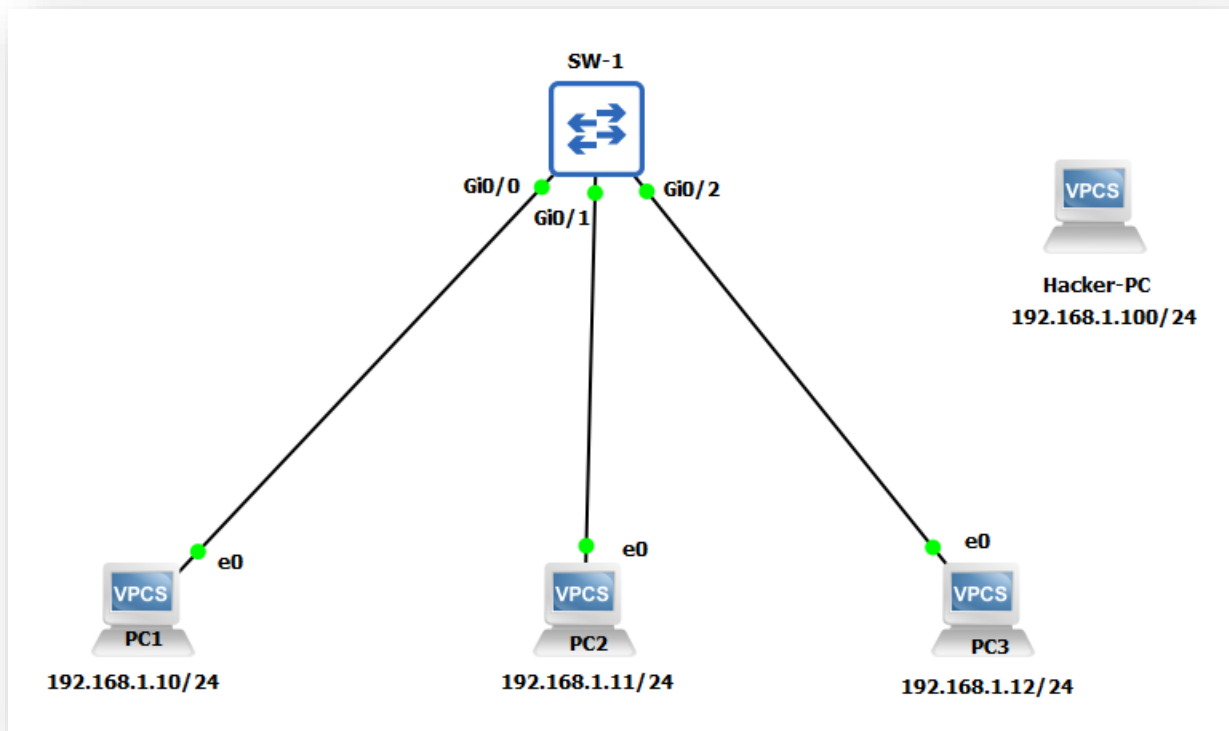


Switchport Security

**Task**

1. Configure the SW-1 with IP 150.150.150.1 for device management.
2. Assign IP address to hosts.
3. Configure switchport security on each port with different violation actions. Configure Port gi0/0 with static mac of PC1 and port gi0/1 & gi0/2 with dynamic mac binding method.
4. Replace each host with Hacker-PC and observe the violations actions.
5. Enable auto recovery of disabled (shutdown) ports.

Task-1: Configure the SW-1 with IP 150.150.150.1 for device management.**SW-1 Configuration:**

```
Switch#
Switch#config t
Switch(config)#hostname SW1
SW1(config)#no ip domain-lookup
SW1(config)#int vlan1
SW1(config-if)#ip address 192.168.1.1 255.255.255.0
SW1(config-if)#no shut
SW1(config-if)#exit
SW1(config)#exit
SW1#
```

Task-2: Assign IP address to hosts.**PC1> ip 192.168.1.10/24 192.168.1.1**

Checking for duplicate address...

PC1 : 192.168.1.10 255.255.255.0 gateway 192.168.1.1

PC2> ip 192.168.1.11/24 192.168.1.1

Checking for duplicate address...

PC1 : 192.168.1.11 255.255.255.0 gateway 192.168.1.1

PC3> ip 192.168.1.12/24 192.168.1.1

Checking for duplicate address...

PC1 : 192.168.1.12 255.255.255.0 gateway 192.168.1.1

Hacker-PC> ip 192.168.1.100/24 192.168.1.1

Checking for duplicate address...

PC1 : 192.168.1.100 255.255.255.0 gateway 192.168.1.1

✓ Verification & Testing:**SW1#sh int status**

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/0		connected	1	a-full	auto	RJ45
Gi0/1		connected	1	a-full	auto	RJ45
Gi0/2		connected	1	a-full	auto	RJ45
Gi0/3		notconnect	1	a-full	auto	RJ45
Gi1/0		notconnect	1	a-full	auto	RJ45
Gi1/1		notconnect	1	a-full	auto	RJ45
Gi1/2		notconnect	1	a-full	auto	RJ45
Gi1/3		notconnect	1	a-full	auto	RJ45
Gi2/0		notconnect	1	a-full	auto	RJ45
Gi2/1		notconnect	1	a-full	auto	RJ45
Gi2/2		notconnect	1	a-full	auto	RJ45
Gi2/3		notconnect	1	a-full	auto	RJ45
Gi3/0		notconnect	1	a-full	auto	RJ45
Gi3/1		notconnect	1	a-full	auto	RJ45
Gi3/2		notconnect	1	a-full	auto	RJ45
Gi3/3		notconnect	1	a-full	auto	RJ45

SW1#

SW1#sh int gi0/0 switchport | section Administrative Mode**Administrative Mode: dynamic auto****SW1#sh int gi0/1 switchport | section Administrative Mode****Administrative Mode: dynamic auto****SW1#sh int gi0/2 switchport | section Administrative Mode****Administrative Mode: dynamic auto**

Ping PC2 & PC3 from PC1**PC1> ping 192.168.1.11**

84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=3.653 ms
84 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=4.348 ms
84 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=4.958 ms
84 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=5.094 ms
84 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=4.584 ms

PC1> ping 192.168.1.12

84 bytes from 192.168.1.12 icmp_seq=1 ttl=64 time=4.631 ms
84 bytes from 192.168.1.12 icmp_seq=2 ttl=64 time=3.476 ms
84 bytes from 192.168.1.12 icmp_seq=3 ttl=64 time=3.343 ms
84 bytes from 192.168.1.12 icmp_seq=4 ttl=64 time=4.612 ms
84 bytes from 192.168.1.12 icmp_seq=5 ttl=64 time=4.720 ms

Ping PC1 & PC3 from PC2**PC2> ping 192.168.1.10**

84 bytes from 192.168.1.10 icmp_seq=1 ttl=64 time=5.173 ms
84 bytes from 192.168.1.10 icmp_seq=2 ttl=64 time=5.669 ms
84 bytes from 192.168.1.10 icmp_seq=3 ttl=64 time=5.523 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=64 time=4.479 ms
84 bytes from 192.168.1.10 icmp_seq=5 ttl=64 time=4.238 ms

PC2> ping 192.168.1.12

84 bytes from 192.168.1.12 icmp_seq=1 ttl=64 time=5.213 ms
84 bytes from 192.168.1.12 icmp_seq=2 ttl=64 time=4.303 ms
84 bytes from 192.168.1.12 icmp_seq=3 ttl=64 time=4.355 ms
84 bytes from 192.168.1.12 icmp_seq=4 ttl=64 time=4.121 ms
84 bytes from 192.168.1.12 icmp_seq=5 ttl=64 time=4.685 ms

Ping PC1 & PC2 from PC3**PC3> ping 192.168.1.10**

84 bytes from 192.168.1.10 icmp_seq=1 ttl=64 time=4.167 ms
84 bytes from 192.168.1.10 icmp_seq=2 ttl=64 time=3.503 ms
84 bytes from 192.168.1.10 icmp_seq=3 ttl=64 time=3.664 ms
84 bytes from 192.168.1.10 icmp_seq=4 ttl=64 time=3.558 ms
84 bytes from 192.168.1.10 icmp_seq=5 ttl=64 time=3.485 ms

PC3> ping 192.168.1.11

84 bytes from 192.168.1.11 icmp_seq=1 ttl=64 time=4.273 ms
84 bytes from 192.168.1.11 icmp_seq=2 ttl=64 time=4.077 ms
84 bytes from 192.168.1.11 icmp_seq=3 ttl=64 time=4.032 ms
84 bytes from 192.168.1.11 icmp_seq=4 ttl=64 time=4.170 ms
84 bytes from 192.168.1.11 icmp_seq=5 ttl=64 time=4.174 ms

SW1#sh mac address-table

Mac Address Table

Vlan	Mac Address	Type	Ports
1	0050.7966.6800	DYNAMIC	Gi0/0
1	0050.7966.6801	DYNAMIC	Gi0/1
1	0050.7966.6802	DYNAMIC	Gi0/2

Total Mac Addresses for this criterion: 3

Task-3: Configure switchport security on each port with different violation actions

Protect – This mode drops the packets with unknown source mac address. When using this mode, no notification message is sent when this violation occurs.

Restrict – This mode performs the same function as protect, i.e., drops the packets with unknown source mac address. In addition to this, it will generate a log message, increment the counter value and will also send SNMP trap.

Shut down – This mode is mostly preferred as compared to other modes as it shut down the port immediately if unauthorized access is done. It will also generate a log, increment counter value and send a SNMP trap. This port will remain in shut down state until the administrator will perform “no shutdown” command.

```
SW1#config t
SW1(config)#int gi0/0
SW1(config-if)#switchport mode access
SW1(config-if)#switchport port-security
SW1(config-if)#switchport port-security mac-address 00:50:79:66:68:00
SW1(config-if)#switchport port-security maximum 1
SW1(config-if)#switchport port-security violation protect
SW1(config-if)#switchport port-security aging time 10
SW1(config-if)#switchport port-security aging type inactivity
SW1(config-if)#exit
SW1(config)#int gi0/1
SW1(config-if)#switchport mode access
SW1(config-if)#switchport port-security
SW1(config-if)#switchport port-security mac-address sticky
SW1(config-if)#switchport port-security maximum 1
SW1(config-if)#switchport port-security violation restrict
SW1(config-if)#switchport port-security aging time 10
SW1(config-if)#switchport port-security aging type inactivity
SW1(config-if)#exit
SW1(config)#int gi0/2
SW1(config-if)#switchport mode access
SW1(config-if)#switchport port-security
SW1(config-if)#switchport port-security mac-address sticky
```

```
SW1(config-if)#switchport port-security maximum 1
SW1(config-if)#switchport port-security violation shutdown
SW1(config-if)#switchport port-security aging time 10
SW1(config-if)#switchport port-security aging type inactivity
SW1(config-if)#exit
SW1(config)#exit
SW1#
```

✓ Verification & Testing:

SW1#sh int gi0/0 switchport | section Administrative Mode
 Administrative Mode: static access

SW1#sh int gi0/1 switchport | section Administrative Mode
 Administrative Mode: static access

SW1#sh int gi0/2 switchport | section Administrative Mode
 Administrative Mode: static access

SW1#sh port-security

Secure Port	MaxSecureAddr (Count)	CurrentAddr (Count)	SecurityViolation (Count)	Security Action
Gi0/0	1	1	0	Protect
Gi0/1	1	0	0	Restrict
Gi0/2	1	0	0	Shutdown

Total Addresses in System (excluding one mac per port) : 0
 Max Addresses limit in System (excluding one mac per port) : 4096

SW1#sh port-security address

Secure Mac Address Table

Vlan	Mac Address	Type	Ports	Remaining Age (mins)
1	0050.7966.6800	SecureConfigured	Gi0/0	-

Total Addresses in System (excluding one mac per port) : 0
 Max Addresses limit in System (excluding one mac per port) : 4096
 SW1#

SW1#sh port-security interface gi0/0

```
Port Security          : Enabled
Port Status            : Secure-up
Violation Mode          : Protect
Aging Time              : 10 mins
Aging Type              : Inactivity
SecureStatic Address Aging : Disabled
Maximum MAC Addresses   : 1
```

Total MAC Addresses : 1
 Configured MAC Addresses : 1
 Sticky MAC Addresses : 0
Last Source Address:Vlan : 0050.7966.6800:1
 Security Violation Count : 0

SW1#sh port-security interface gi0/1

Port Security : Enabled
 Port Status : Secure-up
Violation Mode : Restrict
 Aging Time : 10 mins
 Aging Type : Inactivity
 SecureStatic Address Aging : Disabled
 Maximum MAC Addresses : 1
 Total MAC Addresses : 1
 Configured MAC Addresses : 0
 Sticky MAC Addresses : 1
Last Source Address:Vlan : 0050.7966.6801:1
 Security Violation Count : 0

SW1#sh port-security interface gi0/2

Port Security : Enabled
 Port Status : Secure-up
Violation Mode : Shutdown
 Aging Time : 10 mins
 Aging Type : Inactivity
 SecureStatic Address Aging : Disabled
 Maximum MAC Addresses : 1
 Total MAC Addresses : 1
 Configured MAC Addresses : 0
 Sticky MAC Addresses : 1
Last Source Address:Vlan : 0050.7966.6802:1
 Security Violation Count : 0

Task-4: Replace each host with Hacker-PC and observe the violations actions.

Test-1: Connect the Hacker-PC on switchport Gi0/0 and ping PC2.

Hacker-PC> sh ip

NAME : Hacker-PC[1]
 IP/MASK : 192.168.1.100/24
 GATEWAY : 192.168.1.1
 DNS :
 MAC : 00:50:79:66:68:03
 LPORT : 10009
 RHOST:PORT : 127.0.0.1:10010
 MTU: : 1500

SW1#debug port-security

All Port Security debugging is on
SW1#

Hacker-PC> ping 192.168.1.11

host (192.168.1.11) not reachable

SW1#

*Jan 2 14:45:51.542: PSECURE: unix_l2_process_psecure_pak: swidb = GigabitEthernet0/0 mac_addr = 0050.7966.6803 vlanid = 1

*Jan 2 14:45:51.542: PSECURE: Violation/duplicate detected upon receiving 0050.7966.6803 on vlan 1: port_num_addrs 1 port_max_addrs 1 vlan_addr_ct 1: vlan_addr_max 1 total_addrs 0: max_total_addrs 4096

SW1#sh port-security interface gi0/0

Port Security : Enabled
Port Status : Secure-up
Violation Mode : Protect
Aging Time : 10 mins
Aging Type : Inactivity
SecureStatic Address Aging : Disabled
Maximum MAC Addresses : 1
Total MAC Addresses : 1
Configured MAC Addresses : 1
Sticky MAC Addresses : 0
Last Source Address:Vlan : 0050.7966.6803:1
Security Violation Count : 0

Test-2: Connect the Hacker-PC on switchport Gi0/1 and ping PC3.**Hacker-PC> ping 192.168.1.12**

host (192.168.1.11) not reachable

SW1#

*Jan 2 14:48:58.058: %PORT_SECURITY-2-PSECURE_VIOLATION: Security violation occurred, caused by MAC address 0050.7966.6803 on port GigabitEthernet0/1.

*Jan 2 14:48:58.058: PSECURE: Security violation, TrapCount:4

*Jan 2 14:48:59.056: PSECURE: psecure_should_be_sticky: Address is sticky on port GigabitEthernet0/1 for vlan 1

*Jan 2 14:48:59.057: PSECURE: unix_l2_process_psecure_pak: swidb = GigabitEthernet0/1 mac_addr = 0050.7966.6803 vlanid = 1

*Jan 2 14:48:59.057: PSECURE: Violation/duplicate detected upon receiving 0050.7966.6803 on vlan 1: port_num_addrs 1 port_max_addrs 1 vlan_addr_ct 1: vlan_addr_max 1 total_addrs 0: max_total_addrs 4096

SW1#sh port-security interface gi0/1

Port Security : Enabled
Port Status : Secure-up
Violation Mode : Restrict
Aging Time : 10 mins
Aging Type : Inactivity
SecureStatic Address Aging : Disabled
Maximum MAC Addresses : 1
Total MAC Addresses : 1

Configured MAC Addresses : 0
 Sticky MAC Addresses : 1
 Last Source Address:Vlan : 0050.7966.6803:1
Security Violation Count : 6

SW1#sh port-security

Secure Port	MaxSecureAddr (Count)	CurrentAddr (Count)	SecurityViolation (Count)	Security Action
Gi0/0	1	1	0	Protect
Gi0/1	1	1	6	Restrict
Gi0/2	1	1	0	Shutdown

Total Addresses in System (excluding one mac per port) : 0
 Max Addresses limit in System (excluding one mac per port) : 4096
 SW1#

Test-3: Connect the Hacker-PC on switchport Gi0/2 and ping PC1.

Hacker-PC> ping 192.168.1.10
host (192.168.1.11) not reachable

SW1#

*Jan 2 14:51:45.206: PSECURE: psecure_should_be_sticky: Address is sticky on port GigabitEthernet0/2 for vlan 1
 *Jan 2 14:51:45.208: PSECURE: unix_l2_process_psecure_pak: swidb = GigabitEthernet0/2 mac_addr = 0050.7966.6803 vlanid = 1
 *Jan 2 14:51:45.209: PSECURE: **Violation/duplicate detected upon receiving 0050.7966.6803 on vlan 1: port_num_addrs 1 port_max_addrs 1 vlan_addr_ct 1: vlan_addr_max 1 total_addrs 0: max_total_addrs 4096**
 *Jan 2 14:51:45.209: PSECURE: psecure_add_addr_check: **Security violation occurred, bring down the interface**
 *Jan 2 14:51:45.209: %PM-4-ERR_DISABLE: **psecure-violation error detected on Gi0/2, putting Gi0/2 in err-disable state**
 *Jan 2 14:51:45.211: PSECURE: psecure_linkchange: Gi0/2 hwidb=0x158CC1F8
 *Jan 2 14:51:45.211: PSECURE: **Link is going down**
 *Jan 2 14:51:45.211: PSECURE: psecure_linkdown_init: Gi0/2 hwidb = 0x158CC1F8
 *Jan 2 14:51:45.211: PSECURE: **psecure_deactivate_port_security: Deactivating port-security feature**
 *Jan 2 14:51:45.211: PSECURE: port_deactivate: port status is 0
 *Jan 2 14:51:45.212: PSECURE: psecure_clear_ha_table: called
 *Jan 2 14:51:45.212: PSECURE: psecure_clear_ha_table: delete 0050.7966.6802 vlan 1
 *Jan 2 14:51:45.212: PSECURE:
 psecure_platform_del_mac_addrs: unimplemented function!
 *Jan 2 14:51:45.213: %PORT_SECURITY-2-PSECURE_VIOLATION: Security violation occurred, caused by MAC address 0050.7966.6803 on port GigabitEthernet0/2.
 *Jan 2 14:51:45.213: PSECURE: Security violation, TrapCount:1
 *Jan 2 14:51:46.220: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down
 *Jan 2 14:51:47.211: %LINK-3-UPDOWN: **Interface GigabitEthernet0/2, changed state to down**
SW1#

SW1#sh port-security interface gi0/2

Port Security : Enabled
 Port Status : Secure-shutdown

Violation Mode : Shutdown
 Aging Time : 10 mins
 Aging Type : Inactivity
 SecureStatic Address Aging : Disabled
 Maximum MAC Addresses : 1
 Total MAC Addresses : 1
 Configured MAC Addresses : 0
 Sticky MAC Addresses : 1
 Last Source Address:Vlan : 0050.7966.6803:1
Security Violation Count : 1

SW1#sh port-security

Secure Port	MaxSecureAddr (Count)	CurrentAddr (Count)	SecurityViolation (Count)	Security Action
Gi0/0	1	1	0	Protect
Gi0/1	1	1	6	Restrict
Gi0/2	1	1	1	Shutdown

Total Addresses in System (excluding one mac per port) : 0
 Max Addresses limit in System (excluding one mac per port) : 4096

SW1#sh int status

Port Name	Status	Vlan	Duplex	Speed	Type
Gi0/0	connected	1	a-full	auto	RJ45
Gi0/1	connected	1	a-full	auto	RJ45
Gi0/2	err-disabled	1	auto	auto	RJ45
Gi0/3	notconnect	1	a-full	auto	RJ45
Gi1/0	notconnect	1	a-full	auto	RJ45
Gi1/1	notconnect	1	a-full	auto	RJ45
Gi1/2	notconnect	1	a-full	auto	RJ45
Gi1/3	notconnect	1	a-full	auto	RJ45
Gi2/0	notconnect	1	a-full	auto	RJ45
Gi2/1	notconnect	1	a-full	auto	RJ45
Gi2/2	notconnect	1	a-full	auto	RJ45
Gi2/3	notconnect	1	a-full	auto	RJ45
Gi3/0	notconnect	1	a-full	auto	RJ45
Gi3/1	notconnect	1	a-full	auto	RJ45
Gi3/2	notconnect	1	a-full	auto	RJ45
Gi3/3	notconnect	1	a-full	auto	RJ45

SW1#

SW1#sh int gi0/2

GigabitEthernet0/2 is down, line protocol is down (err-disabled)

Hardware is iGbE, address is 0c86.8344.3002 (bia 0c86.8344.3002)

MTU 1500 bytes, BW 1000000 Kbit/sec, DLY 10 usec,

reliability 255/255, txload 1/255, rxload 1/255

Encapsulation ARPA, loopback not set

Keepalive set (10 sec)

Auto Duplex, Auto Speed, link type is auto, media type is unknown media type

output flow-control is unsupported, input flow-control is unsupported
 Auto-duplex, Auto-speed, link type is auto, media type is RJ45
 input flow-control is off, output flow-control is unsupported
 ARP type: ARPA, ARP Timeout 04:00:00
 Last input 00:05:49, output 00:05:50, output hang never
 Last clearing of "show interface" counters never
 Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
 Queueing strategy: fifo
 Output queue: 0/0 (size/max)
 5 minute input rate 0 bits/sec, 0 packets/sec
 5 minute output rate 0 bits/sec, 0 packets/sec
 44 packets input, 4006 bytes, 0 no buffer
 Received 4 broadcasts (4 multicasts)
 0 runts, 0 giants, 0 throttles
 0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
 0 watchdog, 4 multicast, 0 pause input
 913 packets output, 70385 bytes, 0 underruns
 0 output errors, 0 collisions, 2 interface resets
 0 unknown protocol drops
 0 babbles, 0 late collision, 0 deferred
 1 lost carrier, 0 no carrier, 0 pause output
 0 output buffer failures, 0 output buffers swapped out

Task-5: Enable auto recovery of disabled (shutdown) ports after 600 sec.

```
SW1#config t
SW1(config)#errdisable recovery cause psecure-violation
SW1(config)#errdisable recovery interval 600
SW1(config)#exit
SW1#
```

✓ Verification & Testing:

SW1#show errdisable recovery

ErrDisable Reason	Timer Status
arp-inspection	Disabled
bpduguard	Disabled
channel-misconfig (STP)	Disabled
dhcp-rate-limit	Disabled
dtp-flap	Disabled
gbic-invalid	Disabled
inline-power	Disabled
l2ptguard	Disabled
link-flap	Disabled
mac-limit	Disabled
link-monitor-failure	Disabled
loopback	Disabled
oam-remote-failure	Disabled
pagp-flap	Disabled
port-mode-failure	Disabled

```

ppoe-ia-rate-limit      Disabled
psecure-violation       Enabled
security-violation      Disabled
sfp-config-mismatch     Disabled
storm-control           Disabled
udld                    Disabled

```

Interface	Errdisable reason	Time left(sec)
unicast-flood	Disabled	
vmps	Disabled	
psp	Disabled	
dual-active-recovery	Disabled	
evc-lite input mapping fa	Disabled	
Recovery command: "clear	Disabled	

Timer interval: 600 seconds

Interfaces that will be enabled at the next timeout:

Interface	Errdisable reason	Time left(sec)
Gi0/2	psecure-violation	20

After 600 sec time expire

SW1#

*Jan 2 14:59:55.592: %LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to up

*Jan 2 14:59:56.594: %LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

*Jan 2 15:00:23.719: PSECURE: psecure_vp_list_fwdchange invoked

SW1#sh int status

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/0		connected	1	a-full	auto	RJ45
Gi0/1		connected	1	a-full	auto	RJ45
Gi0/2		connected	1	a-full	auto	RJ45
Gi0/3		notconnect	1	a-full	auto	RJ45
Gi1/0		notconnect	1	a-full	auto	RJ45
Gi1/1		notconnect	1	a-full	auto	RJ45
Gi1/2		notconnect	1	a-full	auto	RJ45
Gi1/3		notconnect	1	a-full	auto	RJ45
Gi2/0		notconnect	1	a-full	auto	RJ45
Gi2/1		notconnect	1	a-full	auto	RJ45
Gi2/2		notconnect	1	a-full	auto	RJ45
Gi2/3		notconnect	1	a-full	auto	RJ45
Gi3/0		notconnect	1	a-full	auto	RJ45
Gi3/1		notconnect	1	a-full	auto	RJ45
Gi3/2		notconnect	1	a-full	auto	RJ45
Gi3/3		notconnect	1	a-full	auto	RJ45

SW1#

Important Commands:

```
sh int status
sh int gi0/0 switchport | section Administrative Mode
sh mac address-table
sh port-security
sh port-security address
sh port-security interface gi0/0
sh errdisable recovery
```

For more labs, join



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